

Weon Youngbeen

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EDUCATION

Seoul National University — B.S. in Computer Science and Engineering

Mar 2019 – Feb 2027 (*Expected*)

GPA: 3.85 / 4.30 · Major GPA: 3.75 / 4.30

Coursework

Database, Data Structures, Operating Systems, Programming Language, Algorithms, Computer Architecture (In-Progress) Multicore Computing, Systems Programming, Software Formal Verification, Compilers, Computer Modeling

PROJECTS

Linux Kernel Systems Programming

C · Linux Kernel · Tizen OS

- Implemented a `ptree` system call using Linux kernel task structures to traverse process hierarchies and safely transfer process information to user space.
- Designed and implemented a reader-writer synchronization primitive with blocking/wakeup semantics and fairness guarantees to prevent writer starvation.
- Implemented and integrated a Weighted Round Robin scheduling class into the Linux kernel, including task enqueue/dequeue and scheduling behavior on SMP systems.
- Developed and debugged kernel modifications in a Tizen/Linux kernel environment, investigating interactions with existing kernel subsystems.

Database System

Python · SQL · Lark · Berkeley DB

- Implemented a DBMS in Python from scratch, including SQL parsing with Lark and persistent storage using Berkeley DB.
- Implemented schema management and SQL operations including CREATE, DROP, INSERT, DELETE, SELECT, UPDATE, and DESC.
- Implemented primary/foreign key constraints, type and NULL checking, referential integrity, column resolution, and cascading updates on deletion.

LLVM Optimization

C++ · LLVM IR · Git

- Developed custom compiler optimization passes for asynchronous loading using C++ and LLVM IR.
- Integrated and tested optimization passes within the LLVM compilation pipeline.
- Led implementation and development coordination in a 4-person team, including code review and Git-based collaboration.

Computer Architecture

Python · RISC-V · Verilog · FPGA

- Extended a RISC-V processor simulator in Python with additional instructions and BTB-based branch prediction.
- Designed and implemented a small processor on FPGA using Verilog, including an ALU and program counter.
- Built a functional simulator in Excel to validate processor behavior and assist debugging.

Programming Language

OCaml

- Implemented semantic translation from K-- programs to a stack-based machine with explicit stack, memory, environment, command, and continuation states.
- Implemented normal-order reduction, CPS transformation, and simple and let-polymorphic type systems for the M language.

TECHNICAL STACK

Languages: C, Python, Java, OCaml, C++, Scala, SQL, Verilog

Systems & Tools: Linux, Git, LLVM

AWARDS

- Korean Olympiad in Informatics — Bronze Medalist, 2011 & 2017
 - Nexon Young Programming Cup — Finalist, 2016
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CERTIFICATIONS & LANGUAGES

- Certified Investment Manager (투자자산운용사) — Korea Financial Investment Association, 2026
 - New TEPS - 364/600
 - TOEIC - 800/990
 - JLPT N1 — 163/180
 - JPT - 955/990
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INTERESTS

Optimization & Algorithms · Trading Systems · Systems & Infrastructure · Functional Programming · Database Systems
· Low-Level Software

MILITARY SERVICE

Republic of Korea Army — Honorable Discharge (Oct 2024 – Apr 2026)